

Assignment:- 1

Frequency of words using MapReduce

1. Creation of text file

Wordcount.txt file will open in nano

Add your text in txt file -

For example - Hadoop is an open-source framework used for storing and processing large datasets across clusters of computers. It uses a distributed file system (HDFS) and the MapReduce programming model. Designed for scalability and fault tolerance, Hadoop enables efficient data analysis and is widely used in big data applications across industries.

2. Once the text is copy-pasted/ Added
3. Press ctrl + o
4. Then, Enter key
5. Lastly, Press ctrl + x

This will save the txt file and roll-back you to terminal

2. Create Mapper code file

After this command mapper_word_count.py will open in nano,
Write the mapper code for word_count

Following is the code -

```
#!/usr/bin/env python3
import sys

for line in sys.stdin:
    words = line.strip().split()
    for word in words:
        print(f"{word}\t1")
```

In above code -

#!/usr/bin/env python3 is important to add.

Once the py code is complete

2. Press ctrl + o
3. Then, Enter key
4. Lastly, Press ctrl + x

This will save the py file and roll-back you to terminal

3. Create Reducer code file

After this command `reducer_word_count.py` will open in nano,
Write the reducer code for `word_count`

Following is the code -

```
#!/usr/bin/env python3
import sys

current_word = None
current_count = 0

for line in sys.stdin:
    word, count = line.strip().split("\t")
    count = int(count)

    if current_word == word:
        current_count += count
    else:
        if current_word:
            print(f"{current_word}\t{current_count}")
            current_word = word
            current_count = count

if current_word == word:
    print(f"{current_word}\t{current_count}")
```

In above code -

`#!/usr/bin/env python3` is important to add.

Once the py code is complete

Press `ctrl + o`

Then, Enter key

Lastly, Press `ctrl + x`

This will save the py file and roll-back you to terminal

If your command went wrong

Re-correct it. Before running that command make sure you delete the output created using following command -

```
hduser@Ubuntu:~$ hadoop fs -rm -r /output
```

2025-04-14 18:00:47,700 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
Deleted /output

And then give your command -

```
hduser@Ubuntu:~$ hadoop jar /usr/local/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.4.jar -input /mapreduce_word_count/word_count.txt -output /output -mapper /home/hduser/mapper_word_count.py -reducer /home/hduser/reducer_word_count.py
```

Or else you will get following error -

2025-04-14 18:00:19,901 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
2025-04-14 18:00:20,774 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-04-14 18:00:21,121 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-04-14 18:00:21,122 INFO impl.MetricsSystemImpl: JobTracker metrics system started
2025-04-14 18:00:21,163 WARN impl.MetricsSystemImpl: JobTracker metrics system already initialized!
2025-04-14 18:00:21,374 ERROR streaming.StreamJob: Error Launching job : Output directory hdfs://localhost:54310/output already exists
Streaming Command Failed!